

Maximum Drawdown (MDD) Defined, With Formula for Calculation

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What Is a Maximum Drawdown (MDD)?

A maximum drawdown (MDD) is the maximum observed loss from a peak to a trough of a portfolio, before a new peak is attained. Maximum drawdown is an indicator of downside risk over a specified time period.

It can be used both as a stand-alone measure or as an input into other metrics such as "[Return over Maximum Drawdown](#)" and the [Calmar Ratio](#). Maximum Drawdown is expressed in percentage terms.

The Formula for Maximum Drawdown Is

$$\text{MDD} = \text{Peak Value} - \text{Trough Value} - \text{Peak Value}$$

Understanding Maximum Drawdown

Maximum drawdown is a specific measure of [drawdown](#) that looks for the greatest movement from a high point to a low point, before a new peak is achieved. However, it's important to note that it only measures the size of the largest loss, without taking into consideration the frequency of large losses. Because it measures only the largest drawdown, MDD does not indicate how long it took an investor to recover from the loss, or if the investment even recovered at all.

Maximum drawdown (MDD) is an indicator used to assess the relative riskiness of one [stock screening](#) strategy versus another, as it focuses on [capital preservation](#), which is a key concern for most investors. For example, two screening strategies can have the same average outperformance, tracking error, and volatility, but their maximum drawdowns compared to the benchmark can be very different.

A low maximum drawdown is preferred as this indicates that losses from investment were small. If an investment never lost a penny, the maximum drawdown would be zero. The worst possible maximum drawdown would be -100%, meaning the investment is completely worthless.

MDD should be used in the right perspective to derive the maximum benefit from it. In this regard, particular attention should be paid to the time period being considered. For instance, a hypothetical long-only U.S. fund Gamma has been in existence since 2000 and had a maximum drawdown of -30% in the period ending 2010. While this may seem like a huge loss, note that the S&P 500 had plunged more than 55% from its peak in

October 2007 to its trough in March 2009. While other metrics would need to be considered to assess Gamma fund's overall performance, from the viewpoint of MDD, it has outperformed its benchmark by a huge margin.

Key Takeaways

- Maximum drawdown (MDD) is a measure of an asset's largest price drop from a peak to a trough.
- Maximum drawdown is considered to be an indicator of downside risk, with large MDDs suggesting that down movements could be volatile.
- While MDD measures the largest loss, it does not account for the frequency of losses, not the size of any gains.

Example of Maximum Drawdown

Consider an example to understand the concept of maximum drawdown. Assume an investment portfolio has an initial value of \$500,000. The portfolio increases to \$750,000 over a period of time, before plunging to \$400,000 in a ferocious bear market. It then rebounds to \$600,000, before dropping again to \$350,000. Subsequently, it more than doubles to \$800,000. What is the maximum drawdown?

The maximum drawdown in this case is $= \frac{\$750,000 - \$350,000}{\$750,000} = -53.33\%$

Note the following points:

- The initial peak of \$750,000 is used in the MDD calculation. The interim peak of \$600,000 is not used, since it does not represent a new high.
- The new peak of \$800,000 is also not used since the original drawdown began from the \$750,000 peak.
- The MDD calculation takes into consideration the lowest portfolio value (\$350,000 in this case) before a new peak is made, and not just the first drop to \$400,000.